

10. Panel Data and Diff-in-Diffs

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Where are we? Where are we going?

- Where we have found good controls:
 - Units randomized to receive control
 - Units with similar values of covariates
 - Units with opposite value of some instrument
 - At a discontinuity in treatment assignment (will cover in W13)

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 - At a discontinuity in treatment assignment (will cover in W13)
- What if we have repeated measurements of the same units?
- Now there are two possible sources of variation to exploit:
 - Exploit **cross-sectional** variation in treatment.
 - Exploit variation in treatment **within a unit over time** (before/after)



Source: *Chapter 5* of *Mostly Harmless Econometrics (Textbook 1)* by J. Angrist & J. Pischke

1/ Difference in Differences

Minimum Wages (Card & Krueger, 1994)

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 - Neighboring PA stays at \$4.25
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- In 1992, NJ minimum wage increased from \$4.25 to \$5.05
 - Neighboring PA stays at \$4.25
 - We observe employment in both states before and after increase
- Look at eastern PA and NJ fast food restaurants.
 - Similar prices, wages, products, etc.
 - Most likely to be affected by the change.

Minimum Wage Law in PA and NJ



Source: <https://www.alamy.com/stock-photo/pennsylvania-map.html>

Difference in Differences Design

- Basic setup: two groups, two time periods.
 - Pre-period ($t = 0$): neither group is treated.
 - Post-period ($t = 1$): one group is treated, the other group remains untreated.

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 - $G_i = 1$ denote those that are treated at $t = 1$.
 - $G_i = 0$ denote those that are always untreated.
- Treatment status in each period:
 - No treatment in the first period for either group: $D_{i0} = 0$.
 - In treated group: $G_i = 1 \rightsquigarrow D_{i1} = 1$
 - In control group: $G_i = 0 \rightsquigarrow D_{i1} = 0$

	Time period	
	Pre-period ($t = 0$)	Post-period ($t = 1$)
Control group ($G_i = 0$)	$D_{i0} = 0$	$D_{i1} = 0$
Treated group ($G_i = 1$)	$D_{i0} = 0$	$D_{i1} = 1$

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- $Y_{it}(d)$ is the potential outcome under treatment d and time t .

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- Again, the individual causal effect is just $Y_{it}(1) - Y_{it}(0)$.
- Consistency: $Y_{it} = D_{it} \cdot Y_{it}(1) + (1 - D_{it}) \cdot Y_{it}(0)$
 - We observe control PO for all units in the first period: $Y_{i0}(0) = Y_{i0}$
 - In treated group: $G_i = 1 \rightsquigarrow Y_{i1} = Y_{i1}(1)$
 - In control group: $G_i = 0 \rightsquigarrow Y_{i1} = Y_{i1}(0)$

Identification Problem

- Average treatment effect on the treated (i.e., ATT):

$$\begin{aligned}\tau_{\text{ATT}} &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) | G_i = 1] \\ &= \mathbb{E}[Y_{i1}(1) | G_i = 1] - \mathbb{E}[Y_{i1}(0) | G_i = 1] \\ &= \underbrace{\mathbb{E}[Y_{i1} | G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) | G_i = 1]}_{(b)}\end{aligned}$$

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- Part (a) is just a conditional average of observed data $\rightsquigarrow$ identified.
- Part (b) is a counterfactual: what would the average outcome in the treated group have been if it have been in control?

Three Control Strategies

$$\tau_{\text{ATT}} = \mathbb{E}[Y_{i1}(1)|G_i = 1] - \mathbb{E}[Y_{i1}(0)|G_i = 1]$$

	Time period	
	Pre-period ($t = 0$)	Post-period ($t = 1$)
Control group ($G_i = 0$)	$\mathbb{E}[Y_{i0}(0) G_i = 0]$	$\mathbb{E}[Y_{i1}(0) G_i = 0]$
Treated group ($G_i = 1$)	$\mathbb{E}[Y_{i0}(0) G_i = 1]$	$\mathbb{E}[Y_{i1}(1) G_i = 1]$

- **#1: Cross-Sectional Design**

- Assumption: mean independence of treatment.

$$\mathbb{E}[Y_{i1}(0)|G_i = 1] = \mathbb{E}[Y_{i1}(0)|G_i = 0]$$

- Use post-treatment control group as control:

$$\tau_{\text{ATT}} = \mathbb{E}[Y_{i1}|G_i = 1] - \mathbb{E}[Y_{i1}|G_i = 0]$$

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Control group ($G_i = 0$)	$\mathbb{E}[Y_{i0}(0) G_i = 0]$	$\mathbb{E}[Y_{i1}(0) G_i = 0]$
Treated group ($G_i = 1$)	$\mathbb{E}[Y_{i0}(0) G_i = 1]$	$\mathbb{E}[Y_{i1}(1) G_i = 1]$

- **#2: Before-and-After Design**

- Assumption: no trends.

$$\mathbb{E}[Y_{i0}(0)|G_i = 1] = \mathbb{E}[Y_{i1}(0)|G_i = 1]$$

- Use pre-period outcome in treated group:

$$\tau_{\text{ATT}} = \mathbb{E}[Y_{i1}|G_i = 1] - \mathbb{E}[Y_{i0}|G_i = 1]$$

Three Control Strategies

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|G_i = 1] - \mathbb{E}[Y_{i1}(0)|G_i = 1]$$

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	Pre-period ($t = 0$)	Post-period ($t = 1$)
Control group ($G_i = 0$)	$\mathbb{E}[Y_{i0}(0) G_i = 0]$	$\mathbb{E}[Y_{i1}(0) G_i = 0]$
Treated group ($G_i = 1$)	$\mathbb{E}[Y_{i0}(0) G_i = 1]$	$\mathbb{E}[Y_{i1}(1) G_i = 1]$

- **#3: Difference-in-Differences**

- Assumption: parallel trends.

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 1]$$

- Use pre-period treated outcome plus trend in control group:

$$\begin{aligned}\tau_{ATT} = & (\mathbb{E}[Y_{i1}|G_i = 1] - \mathbb{E}[Y_{i0}|G_i = 1]) \\ & - (\mathbb{E}[Y_{i1}|G_i = 0] - \mathbb{E}[Y_{i0}|G_i = 0])\end{aligned}$$

Parallel Trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 1]$$

- Key assumption of difference-in-differences: **parallel trends**

Parallel Trends

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- Key assumption of difference-in-differences: **parallel trends**
- Interpretation:
 - Secular trends in the control group is a good proxy for how the treated group would have changed over time without treatment.
 - p.s., a secular trend = a general trend built up over the years.

Parallel Trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 1]$$

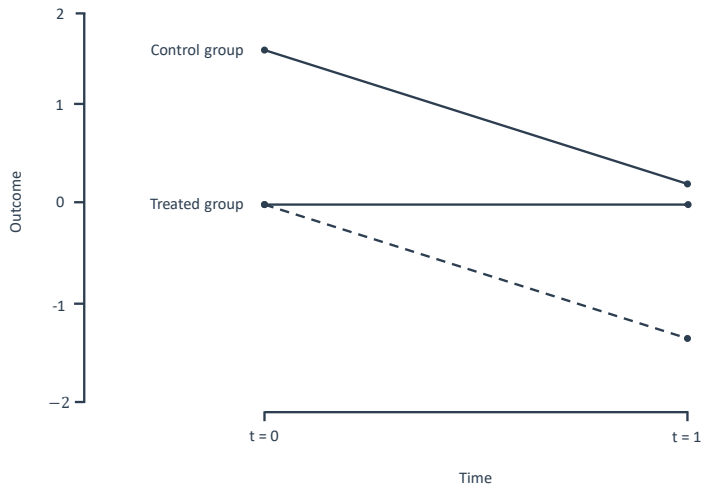
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- Why is this weaker than other assumptions?
 - It allows for time-constant unmeasured confounding b/w Y_{it} and G_i .
 - It allows for (common) secular trends in the outcome over time.

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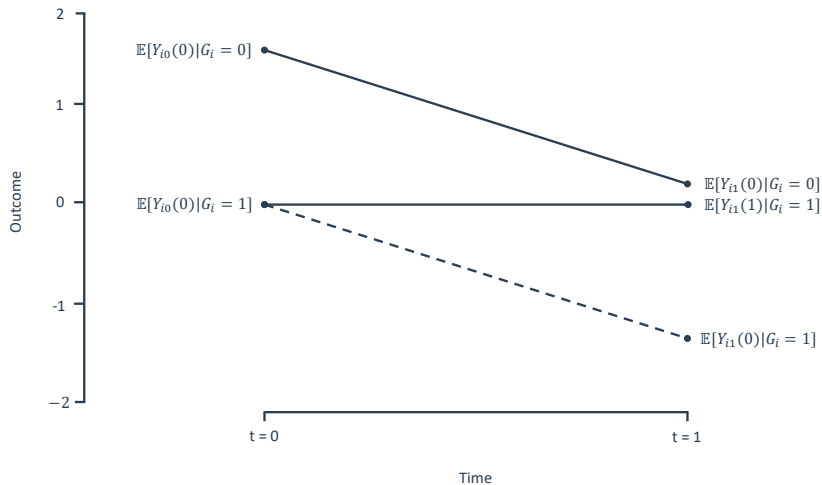
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- Why is this weaker than other assumptions?
 - It allows for time-constant unmeasured confounding b/w Y_{it} and G_i .
 - It allows for (common) secular trends in the outcome over time.
- Not invariant to non-linear transformations:
 - Parallel trends for Y_{it} implies non-parallel trends for $\log(Y_{it})$ vice versa.

Parallel Trends in a Graph



Parallel Trends in a Graph



Identification

- Identification result:

$$\begin{aligned}\tau_{\text{ATT}} = & (\mathbb{E}[Y_{i1}|G_i = 1] - \mathbb{E}[Y_{i0}|G_i = 1]) \\ & - (\mathbb{E}[Y_{i1}|G_i = 0] - \mathbb{E}[Y_{i0}|G_i = 0])\end{aligned}$$

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- Threat to identification: non-parallel trends
 - **Unmeasured time-variant confounders**
 - **Ashenfelter's dip**: empirical finding that people who enroll in job training programs see their earnings decline prior to that training ([link](#)).

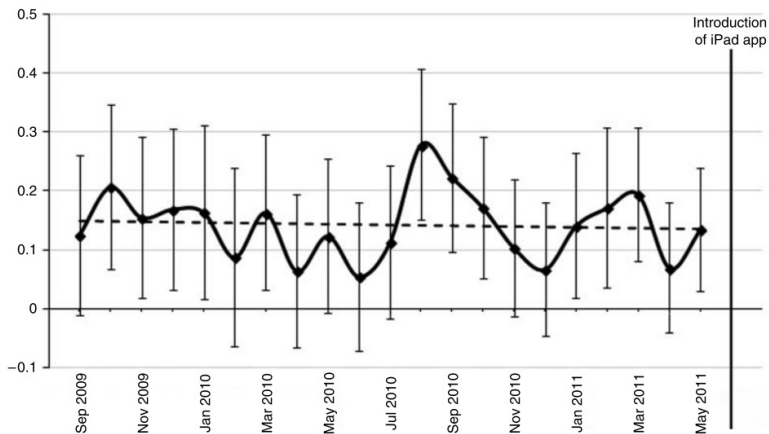
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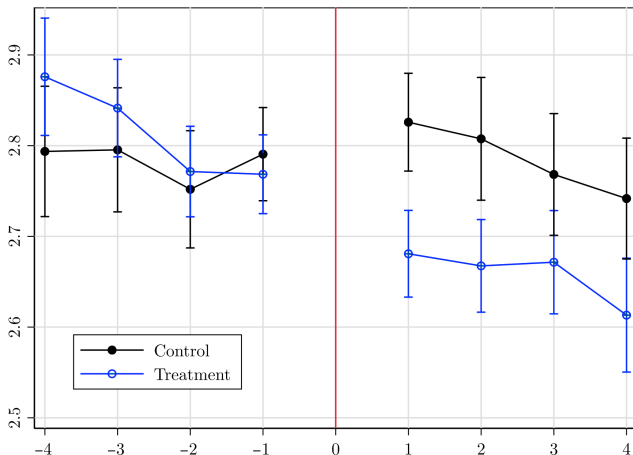
- Threat to identification: non-parallel trends
 - **Unmeasured time-variant confounders**
 - **Ashenfelter's dip**: empirical finding that people who enroll in job training programs see their earnings decline prior to that training ([link](#)).
- Falsification test: check pre-treatment parallel trends.
 - Does's imply parallel trends hold for the post-period, however!

Checking Parallel Trends (Xu et al., 2017)



Source: Xu, Kaiquan, Jason Chan, Anindya Ghose, and Sang Pil Han. "Battle of the channels: The impact of tablets on digital commerce." *Management Science* 63, no. 5 (2017): 1469-1492.

Checking Parallel Trends (Kretschmer & Peukert, 2020)



Source: Kretschmer, Tobias, and Christian Peukert. "Video killed the radio star? Online music videos and recorded music sales." *Information Systems Research* 31, no. 3 (2020): 776-800.

Estimation

- Estimation with panel data:

$$\widehat{\tau}_{\text{ATT}} = \underbrace{\frac{1}{n_1} \sum_{i=1}^n G_i \{Y_{i1} - Y_{i0}\}}_{\text{average trend in treated group}} - \underbrace{\frac{1}{n_0} \sum_{i=1}^n (1 - G_i) \{Y_{i1} - Y_{i0}\}}_{\text{average trend in control group}}$$

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 - Regress $\Delta Y_i = Y_{i1} - Y_{i0}$ on G_i .
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- Regression implementation:
 - Regress $\Delta Y_i = Y_{i1} - Y_{i0}$ on G_i .
 - Use (cluster) robust SEs.
- Also possible to use DID on repeated cross sections.

DID and Linear Two-Way Fixed Effects

- Linear two-way (group and time) fixed effects model:

$$Y_{it} = \gamma G_i + \beta t + \tau D_{it} + \varepsilon_{it}$$

- Fixed effect for group and time.
- D_{it} is treatment status for unit i at time t (i.e., $G_i \times t$).
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- Be sure to cluster by unit (or level of treatment assignment).
- Coefficient on D_{it} equivalent to DID estimation.
- Only holds for the 2 group, 2 period case!
 - Large new literature on interpretation of TWFE in more general cases.
 - Basically, TWFE is an odd weighted average of DID effects with sometimes negative weights.

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 - *We will discuss more when we cover matching methods (e.g., PSM, CEM)!*

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 - *We will discuss more when we cover matching methods (e.g., PSM, CEM)!*
- Beyond covariate matching, **SC/SDiD** families also handle fragile PT (see handout 10(b)).

Example R Codes

```
1 > minwage <- read_csv(url("https://bit.ly/3QZmodH")) # Don't forget to load tidyverse!
2 > minwage <- minwage |>
3   mutate(
4     state = if_else(location == "PA", "PA", "NJ"),
5     treated = if_else(location == "PA", 0, 1) ## PA is control
6   ); head(minwage)
7
8 # A tibble: 6 × 10
9   chain      location wageBefore wageAfter fullBefore fullAfter partBefore partAfter state treated
10  <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl> <chr>    <dbl>
11 1 wendys     PA              5      5.25      20       0       20      36 PA      0
12 2 wendys     PA             5.5      4.75       6      28      26       3 PA      0
13 3 burgerking PA              5      4.75      50      15      35      18 PA      0
14 4 burgerking PA              5       5       10      26      17       9 PA      0
15 5 kfc       PA             5.25      5         2       3       8      12 PA      0
16 6 kfc       PA              5       5         2       2      10       9 PA      0
17
18 > minwage |>
19   count(location)
20 # A tibble: 5 × 2
21   location      n
22  <chr>    <int>
23 1 PA          67
24 2 centralNJ   45
25 3 northNJ    146
26 4 shoreNJ     33
27 5 southNJ     67
```

Example R Codes

```
1 # Simple DID
2 > minwage <- minwage |>
3   mutate(part_prop_after = partAfter / (fullAfter + partAfter),
4           part_prop_before = partBefore / (fullBefore + partBefore),
5           trend = part_prop_after - part_prop_before
6   )
7 > n1 = sum(minwage$treated); n0 = sum(1 - minwage$treated); n1; n0
8 [1] 291
9 [1] 67
10 > did_estimate <- mean(minwage$trend[minwage$treated==1]) -
11   mean(minwage$trend[minwage$treated==0]); did_estimate # Our DID estimate is -0.06155831
12 [1] -0.06155831
13
14 # Regression Implementation
15 > require(estimatr)
16 > estimatr::lm_est <- lm_robust(trend ~ treated, data = minwage, se_type = "HC2")
17 > cat("did estimate: ", lm_est$coefficients[2], "; robust SE estimate:", lm_est$std.error[2])
18 did estimate: -0.06155831 ; robust SE estimate: 0.04551265
```


Example R Codes

```
1 # Two-Way Fixed Effect (TWFE) Regression
2 > require(fixest)
3 > minwage_long <- minwage |>
4   pivot_longer(
5     cols = c(part_prop_after, part_prop_before),
6     names_to = "post",
7     values_to = "part_prop"
8   ) |>
9   mutate(post = if_else(post == "part_prop_before", 0, 1),
10     year = if_else(post == 0, 1991, 1992))
11
12 > twfe_est <- feols(part_prop ~ treated:post|state + year, data = minwage_long)
13 > summary(twfe_est, cluster = "chain")
14
15 OLS estimation, Dep. Var.: part_prop
16 Observations: 716
17 Fixed-effects: state: 2, year: 2
18 Standard-errors: Clustered (chain)
19               Estimate Std. Error t value Pr(>|t|)
20 treated:post -0.061558   0.021945 -2.80517 0.067562 .
21 ---
22 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
23 RMSE: 0.240757   Adj. R2: -2.787e-4
24               Within R2: 0.00248
```



On to the Presentations & Discussions!

Contact Information:

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<https://j1yoo.github.io/>



Appendix

Basic Idea Behind Fixed Effects

- Given a panel data (repeated measurements of the same units),

$$\begin{aligned} Y_{it} &= \alpha + \tau D_{it} + \mathbf{X}'_{it} \beta + \varepsilon_{it} \\ &= \alpha + \tau D_{it} + \mathbf{X}'_{it} \beta + \underbrace{\gamma_i + \lambda_t + \nu_{it}}_{\varepsilon_{it} \text{ decomposed}} \end{aligned}$$

- Individual fixed effects (i.e., γ_i): unobserved innate differences between individuals that are time-invariant.
 - Unobserved factors captured by γ_i , e.g.:
 - 1) ability or preferences (at the individual-level),
 - 2) atmosphere or management style (at the firm-level), or
 - 3) brand image or fashion trend (at the product-level).
 - Observed factors? $\rightsquigarrow$ captured with the covariates (i.e., $\mathbf{X}'_{it}$)!
- Time fixed effects (i.e., λ_t): Impacts common to all groups but vary by year (e.g., seasonality)
- Idiosyncratic error term (i.e., ν_{it})

Basic Idea Behind Fixed Effects

- $\rightsquigarrow$ FE model takes the original econometric error term (i.e., ε_{it}) and decomposes it into three pieces (i.e., γ_i , λ_t , and ν_{it})!

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- $\rightsquigarrow$ FE model takes the original econometric error term (i.e., ε_{it}) and decomposes it into three pieces (i.e., γ_i , λ_t , and ν_{it})!
- Alternative to FE specification is random effects (e.g., Wooldridge, 2006) which “ignores” the indiv. & time FE (in cases where it will not create any bias from ignoring it).
 - In this case, the FEs remain as part of the econometric error term.
 - Wooldridge, Jeffrey M. “Introductory econometrics: A modern approach, 3rd.” New York: Thomson 53 (2006).

Semiparametric Estimation w/ Repeated Outcomes

- How to estimate regression DID without strong linearity assumptions?

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- Abadie (2005) derives **weighting estimators** in this setting:

$$\mathbb{E} [Y_i(1) - Y_i(0) | G_i = 1] = \mathbb{E} \left[\frac{(Y_{i1} - Y_{i0})}{\mathbb{P}(G_i = 1)} \cdot \frac{G_i - \mathbb{P}(G_i = 1 | \mathbf{x}_i)}{1 - \mathbb{P}(G_i = 1 | \mathbf{x}_i)} \right]$$

Semiparametric Estimation w/ Repeated Outcomes

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- Abadie (2005) derives **weighting estimators** in this setting:

$$\mathbb{E} [Y_i(1) - Y_i(0) | G_i = 1] = \mathbb{E} \left[\frac{(Y_{i1} - Y_{i0})}{\mathbb{P}(G_i = 1)} \cdot \frac{G_i - \mathbb{P}(G_i = 1 | \mathbf{X}_i)}{1 - \mathbb{P}(G_i = 1 | \mathbf{X}_i)} \right]$$

- Reweights control group to have the same distribution of $\mathbf{X}_i$ as treated group.

Semiparametric Estimation w/ Repeated Outcomes

- How to estimate regression DID without strong linearity assumptions?
- Abadie (2005) derives **weighting estimators** in this setting:

$$\mathbb{E} [Y_i(1) - Y_i(0) | G_i = 1] = \mathbb{E} \left[\frac{(Y_{i1} - Y_{i0})}{\mathbb{P}(G_i = 1)} \cdot \frac{G_i - \mathbb{P}(G_i = 1 | \mathbf{X}_i)}{1 - \mathbb{P}(G_i = 1 | \mathbf{X}_i)} \right]$$

- Reweights control group to have the same distribution of $\mathbf{X}_i$ as treated group.
- Have to estimate the propensity score $\mathbb{P}(G_i = 1 | \mathbf{X}_i)$
 - Issue: possible model misspecification!